BuildMind

AI-Powered Residential House Design and Construction Planning System

PROJECT COST ESTIMATION REPORT

University Project | Sri Lanka | Five-Member Dataset Team

Document detail	Value
Prepared for	University assessment / project proposal
Date	16 October 2026
Planned development period	6 weeks (target); 8 weeks maximum
Currency	Sri Lankan Rupees (LKR)
Document status	Budget estimate - not an invoice or supplier quotation

Prepared by: BuildMind Project Team
Institution / course / student IDs: To be completed before submission

1. Executive Summary

BuildMind is a proposed university software project that accepts residential plot dimensions and user requirements, generates preliminary 2D layouts, displays interactive 3D visualizations, and provides indicative construction cost estimates. The project uses existing student laptops and open-source development tools. A five-member dataset team is assumed for this estimate.

Budget measure	Amount (LKR)
Estimated cash budget for six-week student development	15,000
Estimated imputed student labour value (not payable)	225,000
Total illustrative economic cost (cash + labour value)	240,000
Equal cash contribution per five members	3,000

The LKR 15,000 amount is a planning ceiling, not a commitment to spend every line item. Costs are indicative allowances rather than verified October 2026 supplier quotations.

2. Project Scope and Cost Basis

Included: requirements preparation, dataset research, Angular frontend, backend API integration, Gemini-based requirement interpretation and chatbot, 2D layout prototype, Three.js visualization, basic cost estimation, testing, demonstration deployment and academic documentation.

Excluded: purchase of new computers, staff salaries, commercial architect or engineer approval, cadastral surveys, construction permits, paid production-grade 3D rendering, commercial-scale traffic and long-term support.

3. Detailed Cash Cost Estimate

Expense category	Estimated LKR
Development software (Angular, Three.js, Blender, VS Code, Git)	0
AI API testing credit allowance	2,500
Weather / solar public data API access	0
Backend hosting allowance	2,500
Frontend hosting (eligible free tier)	0
Database / storage (eligible free tier)	0
Optional domain registration allowance	3,000
Dataset research and communications	2,000
Testing / deployment allowance	1,000
Documentation and printing	1,000
Contingency reserve	3,000
TOTAL ESTIMATED CASH BUDGET	15,000

Calculation: 2,500 + 2,500 + 3,000 + 2,000 + 1,000 + 1,000 + 3,000 = LKR 15,000. All zero-cost entries assume compliance with free-tier terms and limits.

4. Budget Justification

AI API (LKR 2,500): capped allowance for prototype prompts, chatbot tests, and retries; not a fixed vendor subscription. Backend hosting (LKR 2,500): short demonstration deployment, potentially unnecessary if hosted locally. Domain (LKR 3,000): optional; can be avoided using a provider subdomain. Research (LKR 2,000): internet/data communications and obtaining source references. Testing (LKR 1,000): integration and demo contingencies. Documentation (LKR 1,000): printing/binding. Reserve (LKR 3,000): unforeseen eligible cash expenses.

5. External Services and Licensing Assumptions

Service	Budget treatment / condition
Gemini API	Usage-dependent; verify selected model pricing and free-tier availability before use.
Open-Meteo	Free non-commercial tier subject to usage conditions; confirm eligibility.
NASA POWER	Public climate/solar datasets; no API allowance.
OpenStreetMap / Nominatim	Public services have acceptable-use policies and rate limits; not a guaranteed production SLA.
GitHub / development libraries	Free/open-source use subject to relevant licenses.
Hosting and database	Free plans may impose quotas or non-commercial restrictions; paid allowance reserved.

The project must not expose AI API keys in the Angular frontend. API requests requiring credentials should be routed through the backend.

6. Five-Member Dataset Work Allocation

Member	Responsibility	Primary deliverable
1	Weather, climate and site conditions	Weather/site CSV and source log
2	Construction materials and cost	Price CSV with quotation dates
3	Room dimensions, doors, windows and objects	Dimension CSV and reference log
4	Room adjacency, functional constraints, preferences	Adjacency/constraint CSV
5	Sri Lankan building rules and architectural styles	Verified rules/style CSV and source log

All five members additionally contribute to integration testing and documentation. Dataset ownership does not imply that a regulatory rule or price has been professionally verified.

7. Labour Effort Valuation (Non-Cash)

Parameter	Assumption
Number of members	5
Project duration	6 weeks
Workload per member	15 hours per week
Total hours	5 x 6 x 15 = 450 hours
Illustrative valuation rate	LKR 500 per hour
Imputed labour value	450 x 500 = LKR 225,000

This labour valuation is for academic economic-cost analysis only. It is not a verified market wage and is not part of the LKR 15,000 cash requirement.

8. Cash Contribution and Cost Controls

Equal sharing assumption: LKR 15,000 / 5 members = LKR 3,000 per member. The team should appoint one treasurer, obtain approval before paid subscriptions, maintain receipts, set an AI usage budget cap, and reconcile actual spending weekly. If domain and paid hosting are unnecessary, the team should avoid these optional expenses.

9. Illustrative Post-Launch Monthly Operating Cost

Operating category	Monthly LKR (planning range)
Frontend hosting	0 - 2,000
Backend hosting	2,000 - 6,000
Database	0 - 8,000
AI API usage	1,000 - 5,000
Storage / backup	0 - 2,000
Domain amortization	250 - 500
TOTAL	3,250 - 23,500

This is a separate future monthly scenario, not an additional six-week development expense. Real operating costs depend on users, traffic, selected services, exchange rates and taxes.

10. Risk and Sensitivity Analysis

Risk	Financial response
AI requests exceed allowance	Limit calls; cache responses; use free quota where eligible.
Free hosting unavailable	Use local demo or spend within backend hosting allowance.
Dataset sourcing requires paid access	Prefer official public references; request approval before purchase.
Foreign exchange / taxes	Use contingency and recheck provider billing before payment.
Schedule extends beyond six weeks	Reforecast hosting/API and document revised approval.

11. Source and Verification Register

Official service pages to verify at procurement time: Google Gemini API pricing (https://ai.google.dev/gemini-api/docs/pricing); Open-Meteo terms (https://open-meteo.com/en/terms); NASA POWER (https://power.larc.nasa.gov/); Nominatim usage policy (https://operations.osmfoundation.org/policies/nominatim/); GitHub pricing (https://github.com/pricing). Prices and free-tier conditions may change. No live quotation was obtained for the LKR allowances in this report.

12. Conclusion

The recommended estimated cash budget for the six-week BuildMind student prototype is LKR 15,000, equivalent to LKR 3,000 per member under equal sharing. The illustrative non-cash student labour valuation is LKR 225,000. The combined illustrative economic project cost is therefore LKR 240,000. This report should be updated with dated quotations, actual invoices and university-specific requirements before final financial approval.

Approval / verification

Project supervisor: _____ Date: _____

Team representative: _____ Date: _____